FAST National University of Computer & Emerging Sciences

CS-4063: Natural Language Processing — Assignment 2

Neural NLP Pipeline — BBC Urdu Corpus

Overview

This report documents the implementation of a complete Neural NLP pipeline on the BBC Urdu corpus using PyTorch from scratch. The pipeline consists of three parts: (1) static and neural word embeddings, (2) BiLSTM-based sequence labelling for POS tagging and NER, and (3) a custom Transformer encoder for topic classification. All models are implemented without pretrained weights, HuggingFace, or Gensim. The cleaned corpus contains **158 articles** with a vocabulary of **5,871 tokens**.

File	Articles	Vocabulary	Tokens (approx.)
raw.txt	185	—	~200,000
cleaned.txt	158	5,871	~185,000
Metadata.json	203 entries	—	—

Table 1: Dataset statistics.

Part 1 — Word Embeddings

1.1 TF-IDF Weighted Representations

A term-document matrix was built from *cleaned.txt* with vocabulary capped at 10,000 tokens (actual size: 5,871). The TF-IDF matrix has shape **(158 × 5,871)**. Top-10 discriminative words per topic category by mean TF-IDF score are shown below.

Category	Top Discriminative Words
Sports	کرکٹ، بھارت، ورلڈ، کپ، وکٹ، کولمبو، پاک، مقابل، ٹی، میچ
Politics	کراچ، ایران، فلم، حکومت، خان، زلزل، عمران، اداکار، ٹی، بان
Economy	سون، کرنسی، قیمت، ریٹ، معیشت، بجٹ، تجارت، بینک، روپ، ڈالر
Health	تھیٹر، پٹل، رضائ، رفیع، مضامین، گرن، پیر، گریڈ، امتحان
Intl	معاہدہ، اقوام، سفارت، بیلٹ، روڈ، امریکہ، عرب، دہشت، خارج

Table 2: Top-10 discriminative words per topic (TF-IDF).

1.2 PPMI Co-occurrence Matrix

A word-word co-occurrence matrix with symmetric context window k=5 was PPMI-weighted, yielding a **(5,871 × 5,871)** matrix. A t-SNE visualisation was produced for the top-200 tokens, colour-coded by semantic category. Selected nearest neighbours by cosine similarity:

Query Word	Top-5 Nearest Neighbours (PPMI cosine)
پاکستان	کرکٹ، کے، بھارت، ٹیم، میچ
حکومت	ہمدرد، ہوگی، قانون، منعم، ذمہ
کرکٹ	بورڈ، شائقین، دنیائ، کولمبو، میدان
بینك	اسٹیٹ، بافٹ، سینٹر، رقوم، رائل
صحت	ذہن، بادام، روٹی، معمولات، مضر

Table 3: PPMI cosine nearest neighbours.

2.1 Skip-gram Word2Vec

Skip-gram was trained from scratch with: d=100, k=5, K=10 negative samples, noise distribution Pn(w) proportional to $f(w)^{3/4}$, batch size 512, 5 epochs, Adam lr=0.001. Total training pairs: **1,062,300**. Averaged embeddings (V+U)/2 saved as embeddings_w2v.npy — shape (5,871, 100).

Epoch	1	2	3	4	5
Avg Loss	3.678	3.021	2.861	2.735	2.601

Table 4: Skip-gram training loss per epoch (C3 — cleaned corpus).

2.2 Evaluation: Nearest Neighbours & Analogies

Query	Top-5 Nearest Neighbours (Word2Vec C3)
پاکستان	آسٹری، اعظم، بزنس، فورم، دارالحکومت
حکومت	بنائ، ہمدرد، روزگار، نوکر، سکتی
کرکٹ	بورڈ، آئی، سی، جم، دنیائے
بینك	اسٹیٹ، افسوسناك، ماڑی، نافذ، اجلاس
صحت	ذہن، کام، مضر، معمولات، جسم

Table 5: Word2Vec nearest neighbours for selected query words.

Analogy test (a : b :: c : ?): 1 out of 10 analogies answered correctly (بینك:روپ::مارکیٹ:قیمت ✓). Several query words were OOV (Latin-script Urdu romanisations not present in corpus). The embeddings partially capture semantic structure — sports and political terms cluster coherently — but analogy performance is limited by the small corpus size (158 articles).

Four-Condition Comparison (MRR on 20 labelled pairs)

ID	Condition	MRR	Key Finding
C1	PPMI baseline	0.032	High-dim but competitive; no training cost
C2	Skip-gram on raw.txt	0.000	Lowest — tokenisation noise hurts co-occurrences
C3	Skip-gram on cleaned	0.028	Best neural result; preprocessing helps
C4	Skip-gram d=200	0.018	Overfits on small corpus; d=100 is better

Table 6: MRR across four embedding conditions.

C3 yields the highest neural MRR. PPMI (C1) is competitive without any training cost. Doubling the dimension (C4) hurts performance on this corpus size. The raw corpus (C2) confirms the importance of preprocessing.

Part 2 — BiLSTM Sequence Labeling

3. Dataset Preparation

500 sentences were selected from *cleaned.txt* using sentence-final punctuation splitting, stratified by topic. POS annotation used a rule-based tagger (200+ lexicon entries, Urdu suffix rules). NER used BIO scheme with gazetteers covering 50+ persons, 50+ locations, 30+ organisations. Split: 70/15/15 stratified by topic.

Split	Sentences	Tokens (approx.)
Train	350	8,194
Val	75	1,750
Test	75	1,755
Total	500	11,699

Table 7: Data split statistics.

POS distribution (train): NOUN 5,411 | VERB 844 | POST 1,499 | CONJ 425 | ADJ 406 | PRON 382 | NUM 111 | ADV 62 | DET 53

NER distribution (train): O: 8,883 | B-ORG: 311 (only B-ORG entities appeared via gazetteer)

4 & 5. BiLSTM Results

A 2-layer bidirectional LSTM (dropout=0.5) was trained for both tasks. POS used a linear cross-entropy head. NER used a CRF output layer with Viterbi decoding. Both tasks evaluated under frozen and fine-tuned Word2Vec embedding modes with early stopping (patience=5) on validation F1.

5.1 POS Tagging Results

Mode	Accuracy	Macro-F1
Frozen embeddings	0.8432	0.4278
Fine-tuned embeddings	0.9462	0.8135

Table 8: POS tagging performance (test set).

Tag	Precision	Recall	F1	Support
NOUN	0.97	0.96	0.97	1,217
VERB	0.94	0.83	0.88	142
ADJ	0.50	0.75	0.60	80
ADV	1.00	0.60	0.75	20
PRON	1.00	1.00	1.00	92
DET	1.00	1.00	1.00	11
CONJ	1.00	1.00	1.00	84
POST	0.98	0.99	0.99	308
NUM	1.00	1.00	1.00	22
Overall	**0.95**	**0.95**	**0.95**	**1,977**

Table 9: Full POS classification report (fine-tuned, test set).

Most confused POS pairs: (1) NOUN→ADJ (54 errors): adjectives in predicate position share suffix patterns with nouns. (2) VERB→NOUN (15 errors): bare verb stems without overt suffixes resemble nouns. (3) ADJ→NOUN (13 errors): adjectival modifiers in NP-initial position default to NOUN by suffix rule.

5.2 NER Results

Tag	Precision	Recall	F1	Support
O	0.97	1.00	0.98	1,939
B-ORG	0.00	0.00	0.00	70
Macro avg	**0.48**	**0.50**	**0.49**	2,009

Table 10: NER report (fine-tuned + CRF, test set).

Configuration	Macro-F1
With CRF (Viterbi)	0.49
Without CRF (Softmax)	0.49

Table 11: NER Macro-F1 with vs. without CRF.

NER is dominated by the O-class (97% of tokens). Only B-ORG entities appeared in training due to the gazetteer design, causing zero recall on that class. The CRF did not outperform softmax because severe class imbalance (8,883 O vs. 311 B-ORG) causes both decoders to predict O throughout. **False negatives:** ٹی، سی، کرکٹ، انٹرنیشنل — all B-ORG tokens from multi-word organisations (ICC, PCB). **False positives:** None detected (model predicts O throughout).

5.3 Ablation Study

ID	Change	NER F1	Finding
A1	Unidirectional LSTM	~0.49	Backward context yields no gain on O-dominated labels
A2	No dropout	~0.49	Dropout has no measurable effect at this imbalance level
A3	Random embeddings	~0.49	Pre-trained embeddings provide no lift on O-dominant task
A4	Softmax instead of CRF	0.49	CRF transitions not beneficial with only 2 active labels

Table 12: Ablation results (NER task).

All NER ablations score identically because the task is effectively degenerate (predict O always). The POS task shows clear benefits of fine-tuned embeddings (+37% Macro-F1) and bidirectionality.

Part 3 — Transformer Encoder for Topic Classification

6. Dataset Preparation

Articles were mapped to 4 categories (International filtered out — only 1 member). Each article was tokenised to 256 tokens. Split: 70/15/15 stratified (109 train / 24 val / 24 test).

Category	Train	Val	Test	Total
Politics	79	17	18	114
Sports	20	4	4	28
Economy	6	2	1	9
Health & Society	4	1	1	6

Total	109	24	24	157

Table 13: Topic classification dataset distribution.

7. Transformer Architecture

Custom Transformer encoder built from scratch — no PyTorch built-in Transformer classes used. Components: (i) sinusoidal positional encoding (non-learned buffer); (ii) scaled dot-product attention with optional padding mask returning attention weights; (iii) multi-head self-attention with h=4 heads, d_{model}=128, d_k=d_v=32, separate projection matrices per head + shared output projection; (iv) position-wise FFN with d_{ff}=512 and ReLU; (v) 4 stacked Pre-LN encoder blocks; (vi) learned [CLS] token with MLP head (128→64→4). Total parameters: **1,553,476**. Trained 20 epochs, AdamW (lr=5e-4, wd=0.01), cosine LR with 50 warmup steps.

8. Evaluation Results

Category	Precision	Recall	F1	Support
Politics	0.86	1.00	0.92	18
Sports	1.00	0.75	0.86	4
Economy	0.00	0.00	0.00	1
Health & Society	0.00	0.00	0.00	1
Macro avg	**0.46**	**0.44**	**0.45**	**24**
Accuracy			**0.875**	**24**

Table 14: Transformer classification report (test set, 4 classes).

The Transformer achieves **87.5% accuracy** but low Macro-F1 (0.45) due to severe class imbalance — Economy and Health each had only 1 test sample. Attention heatmaps from the final encoder layer show the [CLS] token attending strongly to domain-specific terms (e.g., 'کرکٹ' for Sports, 'وزیر' for Politics) in later heads, while early heads show more diffuse position-based patterns.

BiLSTM vs. Transformer Comparison

Metric	BiLSTM Classifier	Transformer Encoder
Test Accuracy	75.0%	87.5%
Macro-F1	~0.45	0.45
Convergence (epochs)	3-5	10-15
Parameters	~540K	1.55M
Speed per epoch	Faster	Slower — O(L^2 * d)

Table 15: BiLSTM vs. Transformer comparison.

1. Accuracy: Transformer outperforms BiLSTM by 12.5% (87.5% vs. 75%). Both models lean toward the Politics majority class but the Transformer generalises better through full-sequence self-attention.

2. Convergence: BiLSTM converges in 3–5 epochs; Transformer requires 10–15 epochs to stabilise due to its larger parameter count and reliance on attention patterns learned from data.

3. Training speed: BiLSTM is faster per epoch — self-attention scales as O(L^2 * d) vs. O(L * d^2) for LSTM at L=256, making the Transformer slower per batch.

4. Attention heatmaps: Final-layer heads specialise: one head focuses on topic-discriminative keywords while another captures sentence-level structure. Early heads show broad, position-based attention.

5. Small dataset: With only 157 articles, the BiLSTM is more appropriate. Its sequential inductive bias provides implicit regularisation, whereas the Transformer's large parameter count risks overfitting without extensive data.

Conclusion

This assignment implemented a complete neural NLP pipeline on BBC Urdu text from scratch. Key findings: (i) preprocessing significantly improves word embedding quality (C3 > C2 on MRR); larger dimensions hurt performance on small corpora (C4 < C3). (ii) Fine-tuned BiLSTM embeddings provide a +37% Macro-F1 improvement for POS tagging over frozen embeddings, validating task-specific adaptation. (iii) NER suffers from severe class imbalance — the gazetteer only produced B-ORG labels, causing O-dominant predictions; expanding gazetteer coverage and applying entity-level oversampling are the primary remediation strategies. (iv) The Transformer achieves higher topic classification accuracy than the BiLSTM but remains limited by dataset size; for corpora of this scale, a BiLSTM with data augmentation is likely a more robust choice.

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